Spacetime Sentinels

A game of low-overhead error detection with spacetime codes, scored entirely with Qiskit

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The question

- How do we get students to **feel** why fault tolerance is hard?
- Our brief: a game built on **error detection with spacetime codes** (`qiskit-paulice`).
- Rules must be the physics. Scores must come from Qiskit, never from hand-set values.

Detection (us)
flag what the checks see, discard those shots. Nothing is repaired.

Mitigation
fix the *estimate* afterwards (ZNE, PEC, twirling, readout).

Correction
logical qubits, repeated stabilizers, a decoder that undoes errors.

A spacetime check in one picture

- One ancilla, one target qubit.
- Controlled X, Y or Z at a few moments in time.
- Syndrome = ancilla bit $\oplus$ parity of some outputs. Deterministic if nothing went wrong.
- A fault is caught if it anticommutes with the check's **flow** at that point.

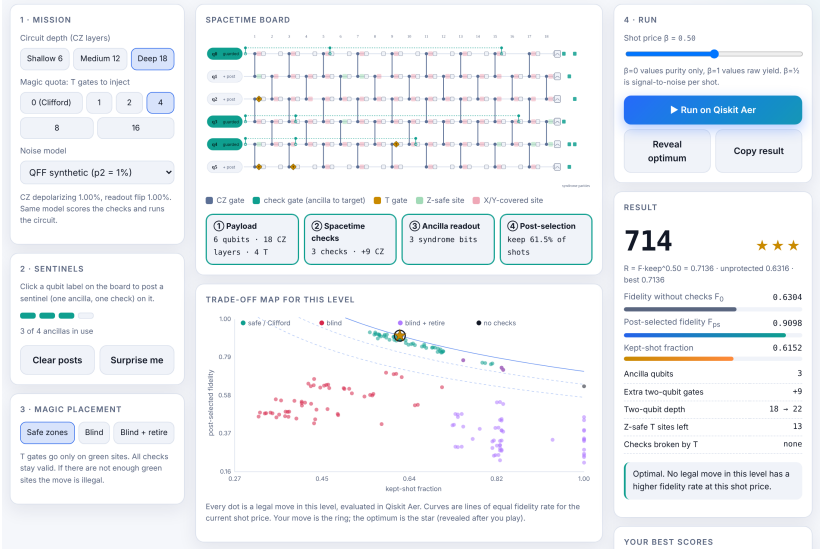
The catch: T gates

$TZT^\dagger = Z$ but $TXT^\dagger = \frac{X+Y}{\sqrt{2}}$

A T on a wire where some check acts as X/Y breaks that check.

On IBM hardware T is a virtual R_Z: **free in noise**, not free for the code.

The game


1 · MISSION
Circuit depth (CZ layers)
Shallow 6
Medium 12
Deep 18
Magic quota: T gates to inject
0 (Clifford)
1
2
4
8
16
Noise model
QFF synthetic (p2 = 1%)
CZ depolarizing 1.00%, readout flip 1.00%. Same model scores the checks and runs the circuit.
2 · SENTINELS
Click a qubit label on the board to post a sentinel (one ancilla, one check) on it.
3 of 4 ancillas in use
Clear posts
Surprise me
3 · MAGIC PLACEMENT
Safe zones
Blind
Blind + retire
T gates go only on green sites. All checks stay valid. If there are not enough green sites the move is illegal.
SPACETIME BOARD
CZ gate
check gate (ancilla to target)
T gate
Z-safe site
X/Y-covered site
① Payload
6 qubits · 18 CZ layers · 4 T
② Spacetime checks
3 checks · +9 CZ
③ Ancilla readout
3 syndrome bits
④ Post-selection
keep 61.5% of shots
TRADE-OFF MAP FOR THIS LEVEL
safe / Clifford
blind
blind + retire
no checks
post-selected fidelity
kept-shot fraction
1.00
0.79
0.58
0.37
0.16
0.27
0.45
0.64
0.82
1.00
Every dot is a legal move in this level, evaluated in Qiskit Aer. Curves are lines of equal fidelity rate for the current shot price. Your move is the ring; the optimum is the star (revealed after you play).
4 · RUN
Shot price β = 0.50
β=0 values purity only, β=1 values raw yield. β=½ is signal-to-noise per shot.
▶ Run on Qiskit Aer
Reveal optimum
Copy result
RESULT
714
★★★
R = F·keep^0.50 = 0.7136 · unprotected 0.6316 · best 0.7136
Fidelity without checks F0
0.6304
Post-selected fidelity Fps
0.9098
Kept-shot fraction
0.6152
Ancilla qubits
3
Extra two-qubit gates
+9
Two-qubit depth
18 → 22
Z-safe T sites left
13
Checks broken by T
none
Optimal. No legal move in this level has a higher fidelity rate at this shot price.
YOUR BEST SCORES

Every rule is a physics decision

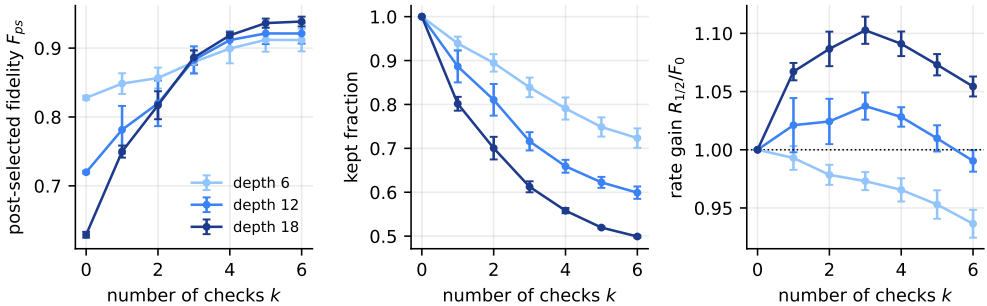
Rule	Physics (and the Qiskit call behind it)
Post a sentinel (budget 4)	one ancilla + one spacetime check, `add_pauli_checks`
Green / red sites	every check acts as I/Z there or not, `uncovered_paulis`
Magic quota	number of T gates in the payload
Safe / Blind / Retire	doping policy; broken checks detected by noiseless statevector
Run	payload $\rightarrow$ checks $\rightarrow$ ancilla readout $\rightarrow$ post-selection, Aer density matrix
Score 1000 $F_{ps}\kappa^{\beta}$	fidelity rate; $\beta = \frac{1}{2}$ is SNR per $\sqrt{\text{shot}}$

5,412 legal moves, every one evaluated exactly with Qiskit Aer.

Why $R = F_{ps}\sqrt{\kappa}$?

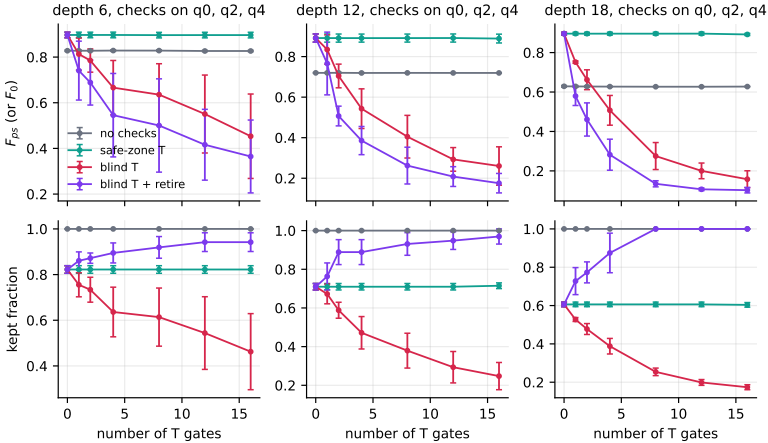
- Residual error damps an observable by F_{ps}; we keep κN of N shots.
- Rescaled estimate has error $\sim \sigma/(F_{ps}\sqrt{\kappa N})$.
- Shots needed $\propto 1/(F_{ps}^2\kappa) = R_{1/2}^{-2}$: the overhead of **detection then mitigation**.
- No checks: $R = F_0$. Gain $G = R/F_0 > 1$ means the sentinels paid for themselves.

Finding 1: checks only pay above a break-even depth



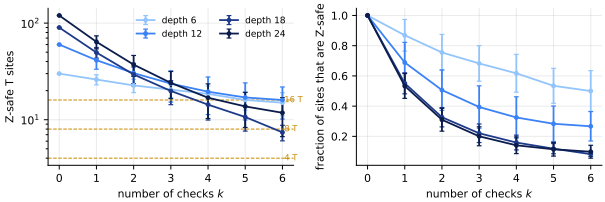
Depth 6: every check lowers R (one check: $G = 0.993$). Best gain at depth 12: 1.037 ($k = 3$); depth 18: 1.103 ($k = 3$). Five random payloads.

Finding 2: T count against check placement



Depth 18, 16 T gates: safe zones $F_{ps} = 0.892$, blind 0.158, blind + retire 0.102, no checks 0.628.

Finding 3: protection and magic compete for spacetime



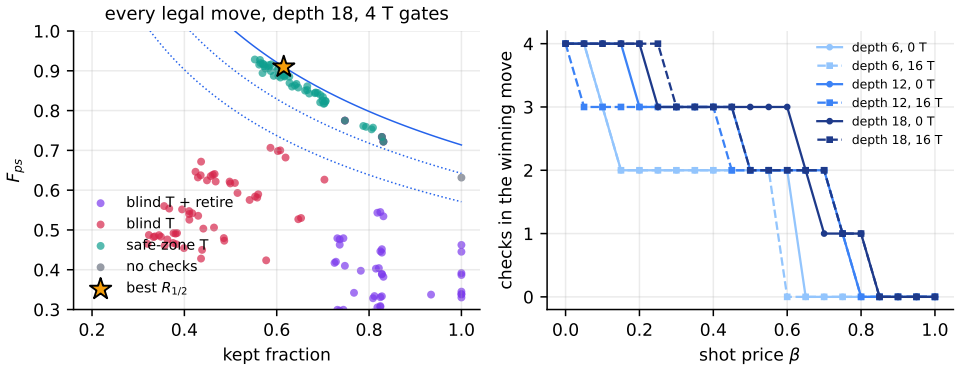
Z-safe T sites at depth 18:
0 checks: 90.0 of 90
2 checks: 29.4
4 checks: 14.3
6 checks: 7.4

Every check you add removes places where magic can go. 25 moves in the game are illegal for exactly this reason.

Finding 4: a broken check is worse than no check

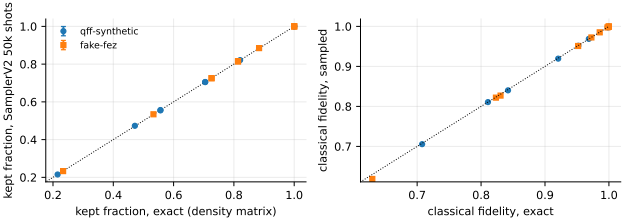
- Blind T placement broke on average 2.4 of 3 checks (depth 18, 4 T).
- Listening to a broken check throws away good shots: it fires on perfect runs.
- **Ignoring it does not help**: its gates still entangle the ancilla with a non-stabilizer operator, a partial measurement that destroys coherence.
- Depth 18, 4 T: blind + retire $F_{ps} = 0.282$ vs. no checks $F_0 = 0.627$.

The best trade-off and why it wins



Deep, 4 T: best move q0,q3,q4 with safe placement, $F_0 = 0.632 \to F_{ps} = 0.910$ at $\kappa = 0.615$, gain $\times 1.130$.

Can you trust the numbers?



- Exact density matrix vs. `SamplerV2`, 50k shots, 20 configurations: max deviation 1.4σ in kept fraction.
- Stabilizer-group identity equals statevector overlap.
- Noiseless runs: valid checks never fire.
- Five random payloads, two noise models.

Qiskit inside

- `qiskit-paulice`: `add_pauli_checks`, `NoiseModel`, `NoiseModel.from_backend(FakeFez)`, `get_postselection_method`, `uncovered_paulis`.
- `qiskit.quantum_info`: `StabilizerState`, `Clifford`, `Statevector`, `hellinger_fidelity`.
- Qiskit Aer: density-matrix simulator, `SamplerV2` primitive, GPU switch.
- Reproducible notebook, tests in CI, one-command rebuild, Slurm scripts.

Honest limitations

- 6 data qubits and ≤ 4 checks so every move is exact. Larger circuits may differ.
- Uniform Pauli noise, no idling noise (as in the current check picker).
- Coherent syndrome parity in simulation (identical statistics to classical XOR).
- Web game plays back pre-computed Qiskit results; live play via `sentinels play`.

Protection and magic compete for the same spacetime.

Play it: `https://OWNER.github.io/spacetime-sentinels`

Code, report and data: `github.com/OWNER/spacetime-sentinels`